

Markov Chains

Dr Feng Chen¹

School of Mathematics & Statistics, UNSW Sydney

¹Slides based on those prepared by previous lecturers of the course, Prof Pierre Del Moral and Dr Leung Chan

Markov Chains

Markov property, Markov transition

Limiting distribution

General state space models

Topological aspects of Markov chains

Markov chains

Definition

Markov chain: $X_n, n \in \mathbb{N}$, taking values in state space S for which the conditional distribution of the random state X_{n+1} w.r.t. to its past and present $(X_0, \dots, X_n)$ depends only on the present state X_n ,

$$\mathbb{P}(X_{n+1} \in dx \mid (X_0, \dots, X_n)) = \mathbb{P}(X_{n+1} \in dx \mid X_n)$$

This property is called the **Markov property**.

The Markov chain is called homogenous whenever the transition probabilities do not depend on the time parameter n .

For any bounded function f on S ,

$$\mathbb{E}(f(X_{n+1}) \mid (X_0, \dots, X_n)) = \mathbb{E}(f(X_{n+1}) \mid X_n)$$

Markov transitions

We focus first on chains with discrete state space S .

Markov property:

$$\mathbb{P}(X_n = x_n \mid X_0, \dots, X_{n-2}, X_{n-1}) = \mathbb{P}(X_n = x_n \mid X_{n-1})$$

Markov transitions (transition kernels) can be identified with a matrix:

$$\forall x, y \in S \quad \mathbb{P}(X_{n+1} = y \mid X_n = x) = P(x, y)$$

$$\mathbb{P}((X_0, \dots, X_n) = (x_0, \dots, x_n) \mid X_0 = x_0) = P(x_0, x_1) \cdots P(x_{n-1}, x_n)$$

$$S = \{e_1, \dots, e_d\}$$

$$\mathbf{P}_n = \begin{pmatrix} P_n(e_1, e_1) & \dots & P_n(e_1, e_d) \\ \vdots & \vdots & \vdots \\ P_n(e_d, e_1) & \dots & P_n(e_d, e_d) \end{pmatrix}$$

Some basic notations

$$\forall 1 \leq i \leq d \quad \underbrace{\mathbb{P}(Y = j)}_{=p_Y(j)} = \sum_{1 \leq i \leq d} \underbrace{\mathbb{P}(X = i)}_{=p_X(i)} \underbrace{\mathbb{P}(Y = j | X = i)}_{=P(i,j)}$$
$$\Updownarrow$$

Matrix notation:

$$p_Y = [\mathbb{P}(Y = 1), \dots, \mathbb{P}(Y = d)]$$

$$= \underbrace{[\mathbb{P}(X = 1), \dots, \mathbb{P}(X = d)]}_{=p_X} \times \underbrace{\begin{pmatrix} \mathbb{P}(Y = 1 | X = 1) & \mathbb{P}(Y = 2 | X = 1) & \dots & \mathbb{P}(Y = d | X = 1) \\ \mathbb{P}(Y = 1 | X = 2) & \mathbb{P}(Y = 2 | X = 2) & \dots & \mathbb{P}(Y = d | X = 2) \\ \vdots & \vdots & \ddots & \vdots \\ \mathbb{P}(Y = 1 | X = d) & \mathbb{P}(Y = 2 | X = d) & \dots & \mathbb{P}(Y = d | X = d) \end{pmatrix}}_{\mathbf{P} = (P(i,j))_{i,j}}$$
$$\Updownarrow$$

Matrix synthetic notation:

$$p_Y = p_X \mathbf{P}$$

$$\mathbb{E}(f(Y) \mid X = i) = \sum_{1 \leq j \leq d} \underbrace{\mathbb{P}(Y = j \mid X = i)}_{=P(i,j)} f(j)$$



Matrix notation:

$$\underbrace{\begin{pmatrix} \mathbb{P}(Y=1 \mid X=1) & \mathbb{P}(Y=2 \mid X=1) & \dots & \mathbb{P}(Y=d \mid X=1) \\ \mathbb{P}(Y=1 \mid X=2) & \mathbb{P}(Y=2 \mid X=2) & \dots & \mathbb{P}(Y=d \mid X=2) \\ \vdots & \vdots & \ddots & \vdots \\ \mathbb{P}(Y=1 \mid X=d) & \mathbb{P}(Y=2 \mid X=d) & \dots & \mathbb{P}(Y=d \mid X=d) \end{pmatrix}}_{\mathbf{P}=(P(i,j))_{i,j}} \underbrace{\begin{pmatrix} f(1) \\ f(2) \\ \vdots \\ f(d) \end{pmatrix}}_{=f} = \underbrace{\begin{pmatrix} \mathbb{E}(f(Y) \mid X=1) \\ \mathbb{E}(f(Y) \mid X=2) \\ \vdots \\ \mathbb{E}(f(Y) \mid X=d) \end{pmatrix}}_{=M(f)}$$



Matrix synthetic notation:

$$\mathbb{E}(f(Y) \mid X = i) = M(f)(i)$$

Markov chain = "sequence of r.v."

$$X_0 \rightsquigarrow X_1 \rightsquigarrow \dots \rightsquigarrow X_{n-1} \rightsquigarrow X_n$$

$$\Updownarrow$$

$$\underbrace{\mathbb{P}(X_n = j)}_{=p_{X_n}(j)} = \sum_{1 \leq i \leq d} \underbrace{\mathbb{P}(X_{n-1} = i)}_{=p_{X_{n-1}}(i)} \underbrace{\mathbb{P}(X_n = j \mid X_{n-1} = i)}_{=P_n(i,j)}$$

$$\Updownarrow$$

Matrix synthetic notation:

$$p_{X_n} = p_{X_{n-1}} P_n = \dots = p_{X_0} P_1 P_2 \dots P_n$$

$$\begin{aligned}
\mathbb{E}(\textcolor{red}{f}(\textcolor{blue}{X}_n) \mid X_0 = i) &= \mathbb{E}\left(\overbrace{\mathbb{E}(f(X_n) \mid X_{n-1})}^{M_n(f)(X_{n-1})} \mid X_0 = i\right) \\
&= \mathbb{E}(\textcolor{red}{M}_n(\textcolor{red}{f})(\textcolor{blue}{X}_{n-1}) \mid X_0 = i) \\
&= \mathbb{E}(\textcolor{red}{M}_{n-1}(\textcolor{red}{M}_n(\textcolor{red}{f}))(\textcolor{blue}{X}_{n-2}) \mid X_0 = i) \\
&= \dots \\
&= \mathbb{E}((\textcolor{red}{M}_1 \dots (\textcolor{red}{M}_n(\textcolor{red}{f}))) (\textcolor{blue}{X}_0) \mid X_0 = i) \\
&= (M_1 M_2 \dots M_n)(f)(i)
\end{aligned}$$

Integral operator

(Chapman-Kolmogorov) Transport equation

$$\begin{aligned}\underbrace{\mathbb{P}(X_n = y)}_{\eta_n(y)} &= \sum_{x \in S} \overbrace{\mathbb{P}(X_n = y \mid X_{n-1} = x)}^{P_n(x, y)} \overbrace{\mathbb{P}(X_{n-1} = x)}^{=\eta_{n-1}(x)} \\ \eta_n(y) &= \sum_{x \in S} \eta_{n-1}(x) P_n(x, y) := (\eta_{n-1} \mathbf{P}_n)(y)\end{aligned}$$

$\Downarrow$

$$\eta_n = \eta_{n-1} \mathbf{P}_n = \dots = \eta_0 \mathbf{P}_1 \dots \mathbf{P}_n$$

with

$$\begin{aligned}(\mathbf{P}_1 \dots \mathbf{P}_n)(x_0, x_n) &= \sum_{(x_1, \dots, x_{n-1}) \in S^{n-1}} P_1(x_0, x_1) \dots P_n(x_{n-1}, x_n) \\ &= \mathbb{P}(X_n = x_n \mid X_0 = x_0)\end{aligned}$$

Equilibrium measures

Homogeneous case $\mathbf{P}_n = \mathbf{P}$

$$\eta_n = \eta_{n-1} \mathbf{P}$$

Definition

Invariant distribution / limiting distribution of the Markov chain X_n with Markov transition $\mathbf{P}$

$$\lim_{n \rightarrow \infty} \eta_n = \lim_{n \rightarrow \infty} \text{Law}(X_n) = \eta_\infty$$

$$\eta_\infty \mathbf{P} = \eta_\infty$$

Definition

A Markov transition P satisfying the reversible property

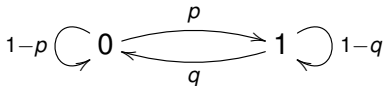
$$\forall x, y \in S \quad \pi(x) P(x, y) = \pi(y) P(y, x)$$

for some measure π is called a **reversible transition** w.r.t. the measure π .

Theorem 1

If a Markov transition $\mathbf{P}$ is reversible transition w.r.t. some probability measure π , then π is an **invariant measure** of the Markov chain with Markov transition $\mathbf{P}$.

Example 2 (2-state model)



Find the invariant measure for the above model.

(Random) dynamical system formulation

Any Markov chain can be described inductively by a (non-unique) recursion:

$$X_{n+1} = F_n(X_n, W_n),$$

where W_n is a sequence of independent r.v.'s taking values in $\mathbb{R}^d$ for some $d \geq 1$ (without loss of generality, W_n can be taken to be *i.i.d.* uniform in $[0, 1]$).

Such formulation often leads to **effective simulation** of Markov chains.

The analysis of Markov chain models is intimately related to the **analysis of dynamical systems with random components**.

Example 3 (Linear Markov chain model)

A **state space model** is described by

$$X_n = F_n(X_{n-1}, W_n) \quad \text{with i.i.d. } W_n \text{ and some initial r.v. } X_0.$$

A special case of such model is the **linear Markov chain model** defined by

$$X_n = A_n X_{n-1} + B_n W_n, \tag{1}$$

where X_0, W_n are R^d -valued independent random variables, and A_n, B_n are $(d \times d)$ -matrices. We further assume that W_n are centered and we denote by $P_n = \text{Cov}(W_n, W_n)$, $n \geq 1$ their covariance matrix; we also set $P_0 = \text{Cov}(X_0, X_0)$. Find an expression for the expectation and variance of X_n .

Example 4

(To be discussed in the tutorial class) Consider the linear Markov chain model above. If we denote by X'_n the solution of the linear system (1) evolving with the same randomness but a different initial variable X'_0 . Assume $(A_n, B_n, \text{Cov}(W_n, W_n)) = (a, b, \sigma^2)$ (Dimension 1 & homogeneous), with $|a| < 1$. Find the following limits:

1. $\lim_{n \rightarrow \infty} |X_n - X'_n|$;
2. $\lim_{n \rightarrow \infty} \mathbb{E}(X_n)$;
3. $\lim_{n \rightarrow \infty} \text{Var}(X_n)$.

Two-state Markov model

$\{0, 1\}$ -valued Markov chain

Choose independent r.v. $W_n = U_n \sim U[0, 1]$, and set

$$\begin{aligned} X_{n+1} &= F_n(X_n, U_n) \\ &:= 1_{[0, M(X_n, 0))}(U_n) \times 0 + 1_{[M(X_n, 0), 1]}(U_n) \times 1 \\ &= 1_{[M(X_n, 0), 1]}(U_n) \end{aligned}$$

with the function

$$F : (x, u) \in \mathcal{S} \times [0, 1] \mapsto F(x, u) := 1_{[M(x, 0), 1]}(u)$$

General state space models

$$\mathbb{E}(f(X_n) \mid X_{n-1} = x_{n-1}) = \int P_n(x_{n-1}, dx_n) f(x_n) := P_n(f)(x_{n-1})$$

(Chapman-Kolmogorov) Transport equation

$$\underbrace{\mathbb{P}(X_n \in dx_n)} = \int_{x_{n-1}} \overbrace{\mathbb{P}(X_n \in dx_n \mid X_{n-1} = x_{n-1})}^{P_n(x_{n-1}, dx_n)} \overbrace{\mathbb{P}(X_{n-1} \in dx_{n-1})}^{=\eta_{n-1}(dx_{n-1})}$$

$$\eta_n(dx_n) = \int_{x_{n-1}} \eta_{n-1}(dx_{n-1}) P_n(x_{n-1}, dx_n) := (\eta_{n-1} \mathbf{P}_n)(dx_n)$$

$$\eta_n = \eta_{n-1} \mathbf{P}_n = \eta_{n-2} \mathbf{P}_{n-1} \mathbf{P}_n = \dots = \eta_0 \mathbf{P}_1 \dots \mathbf{P}_n$$

with

$$\begin{aligned} (\mathbf{P}_1 \dots \mathbf{P}_n)(x_0, dx_n) &= \mathbb{P}(X_n \in dx_n \mid X_0 = x_0) \\ &= \int_{x_1, \dots, x_{n-1}} P_1(x_0, dx_1) \dots P_n(x_{n-1}, dx_n) \\ &= \int_{x_1, \dots, x_{n-1}} (P_1 \dots P_{n-1})(x_0, dx_{n-1}) P_n(x_{n-1}, dx_n) \\ &= \int_{x_1, \dots, x_{n-1}} P_1(x_0, dx_1) (\mathbf{P}_2 \dots \mathbf{P}_n)(x_1, dx_n) \end{aligned}$$

Semigroup (*abbreviated sg*) of the Markov chain

$$\forall 0 \leq p \leq n \quad \mathbf{P}_{p,n} = \mathbf{P}_p \mathbf{P}_{p+1} \dots \mathbf{P}_n$$

$$\forall 0 \leq p \leq q \leq n,$$

$$\mathbf{P}_{p,n} = \mathbf{P}_{p,q} \mathbf{P}_{q,n} \quad \text{and} \quad \eta_n = \eta_p \mathbf{P}_{p,n}$$

Example 5 (Gaussian Markov transitions)

$$P_n(x_{n-1}, dx_n) = \frac{1}{\sqrt{2\pi b_n^2 \sigma_n^2}} \exp \left\{ -\frac{1}{2b_n^2 \sigma_n^2} (x_n - a_n x_{n-1})^2 \right\} dx_n$$

$\Downarrow$

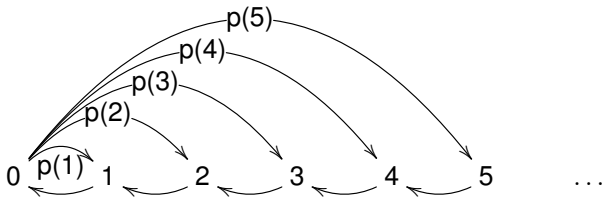
$$X_n = a_n X_{n-1} + b_n W_n, \quad n \geq 1$$

with $(a_n, \sigma_n) \in \mathbb{R}^2$, $b_n \in \mathbb{R} - \{0\}$, and $W_n, n = 1, 2, \dots \stackrel{\text{ind.}}{\sim} N(0, \sigma_n^2)$.
Assume $X_0 \sim N(0, \sigma^2)$, $a_n = a = \sqrt{1 - \epsilon}$ ($\epsilon \in (0, 1]$), $b_n = b = \sqrt{\epsilon}$,
and $\sigma_n = \sigma$. Find the limiting invariant measure of this chain.

Example 6 (Countable state space models)

Markov chain $\in S = \{0, 1, 2, \dots\}$, $p(i)$ a distribution on the set $\{1, 2, \dots\}$ such that $\sum_{i \geq 1} ip(i) < \infty$. Consider the Markov transitions

$$\begin{aligned} P(0, i) &= p(i) & \forall i \geq 1 \\ P(i, i-1) &= 1 & \forall i > 0 \end{aligned} \quad \text{otherwise } P(i, j) = 0.$$



Let I_n be independent r.v. with common law p . Then,

$$X_n = (X_{n-1} - 1) 1_{X_{n-1} \neq 0} + 1_{X_{n-1} = 0} I_n$$

Find the invariant measure of X_n .

Example 7

Let $\mathbf{P}$ be a stochastic matrix on a finite set S . Show that a distribution π is invariant for $\mathbf{P}$ if and only if $\pi(\mathbf{I} - \mathbf{P} + \mathbf{A}) = \mathbf{a}$, where $\mathbf{A} = (a_{ij} : i, j \in S)$ with $a_{ij} = 1$ for all i and j , and $\mathbf{a} = (a_i : i \in S)$ with $a_i = 1$ for all i .

Example 8

Let X be a reversible Markov chain with transition probabilities (p_{ij}) and invariant distribution π . Let C be a non-empty subset of the state space S . Define the Markov chain Y on S by the transition matrix $Q = (q_{ij})$ where

$$q_{ij} = \begin{cases} \beta p_{ij} & \text{if } i \in C \text{ and } j \notin C \\ p_{ij} & \text{otherwise} \end{cases}$$

for $i \neq j$ and β is a constant satisfying $0 < \beta < 1$. The diagonal terms q_{ii} are arranged so that Q is a Markov transition probability matrix.

1. Show that Y is reversible, and find its invariant distribution.
2. Describe the situation in the limit as $\beta \downarrow 0$.

Example 9

1. A random walk on the set $\{0, 1, 2, \dots, b\}$ has transition matrix given by $P_{00} = 1 - \lambda_0$, $P_{bb} = 1 - \mu_b$, $P_{i,i+1} = \lambda_i$ and $P_{i+1,i} = \mu_{i+1}$ for $0 \leq i < b$, where $0 < \lambda_i, \mu_i < 1$ for all i , and $\lambda_i + \mu_i = 1$ for $1 \leq i < b$. Show that this process is reversible in equilibrium.
2. N black balls and N white balls are placed in two urns so that each contains N balls. After each unit of time one ball is selected at random from each urn, and the two balls thus selected are interchanged. Let the number of black balls in the first urn denote the state of the system. Write down the transition matrix of this Markov chain and find the unique stationary distribution. Is the chain reversible in equilibrium?

Markov Chains

Topological aspects of Markov chains

Irreducibility and aperiodicity

Recurrence and transience

Irreducibility and aperiodicity

Definition

Irreducible: All states x and y communicate, i.e.

$\exists n$ s.t. $\mathbf{P}^n(x, y) > 0$

Aperiodic: When the $\gcd \{n : \mathbf{P}^n(x, x) > 0\} = 1$, for any x .
Otherwise the chain is **periodic**.

Any **irreducible aperiodic** (finite) Markov chain has a **unique stationary distribution**.

Lemma 1

For a finite Markov chain, **Aperiodicity** $\oplus$ **irreducibility** $\implies$

$\exists m$ s.t. $\mathbf{P}^m(x, y) > 0 \forall x, y$.

Turn a given Markov chain with transition $\mathbf{P}$ into an aperiodic chain:

$$\mathbf{P}_\epsilon(x, y) = \epsilon \mathbf{1}_x(y) + (1 - \epsilon) \mathbf{P}(x, y)$$

for some $\epsilon \in (0, 1)$.

- ▶ The new chain is **aperiodic**, since $\mathbf{P}_\epsilon(x, x) > 0$.
- ▶ It has the **same invariant measure** as $\mathbf{P}$

$$\pi \mathbf{P} = \pi \implies \pi \mathbf{P}_\epsilon = \pi$$

Definition

The chain with Markov transition $\mathbf{P}_\epsilon$ is called the **ϵ -lazy version** of the Markov chain associated with $\mathbf{P}$.

Sampling independent Bernoulli r.v. $\epsilon \in \{0, 1\}$ with $P_0 = \epsilon$

- ▶ $0 \rightarrow$ stay in the current state.
- ▶ $1 \rightarrow$ move w.r.t. $\mathbf{P}$.

Theorem 10 (Kac's formula)

The **invariant measure** of an **irreducible aperiodic** Markov chain X_n on some finite space S is given by

$$\pi(x) = \frac{1}{\mathbb{E}(T \mid X_0 = x)}$$

where T stands for the **first return time** to x

$$T = \inf \{n \geq 1 : X_n = x\}$$

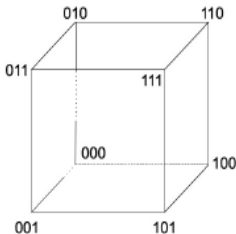
Example 11 (Random walk on hypercubes)

Random walk $X_n = (X_n^i)_{1 \leq i \leq d}$ **on the hypercube** $S = \{0, 1\}^d$ is defined by the **Markov transitions**

$$\mathbf{P}(x, y) = \sum_{1 \leq i \leq d} \mu(i) \mathbf{1}_{x^{(i)}(y)}$$

for some probability measure μ on $\{1, \dots, d\}$, with the mapping $x \mapsto x^{(i)}$

$$\forall k \neq i \quad x^{(i)}(k) = x(k) \quad \text{and} \quad x^{(i)}(i) = 1 - x(i)$$



Example 11 (Random walk on hypercubes)

The number of 1's in the sequence X_n

$$\bar{X}_n := \sum_{1 \leq i \leq d} X_n^i$$

$$\mathbb{E}(f(\bar{X}_{n+1}) \mid X_n) = \left(\sum_{i: X_n^i=1} \mu(i) \right) f(\bar{X}_n - 1) + \left(\sum_{i: X_n^i=0} \mu(i) \right) f(\bar{X}_n + 1)$$

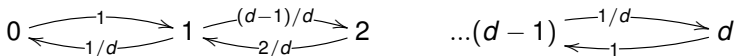
If μ is the uniform probability,

$$\mathbb{E}(f(\bar{X}_{n+1}) \mid X_n) = \frac{\bar{X}_n}{d} f(\bar{X}_n - 1) + \frac{d - \bar{X}_n}{d} f(\bar{X}_n + 1)$$

Example 11 (Random walk on hypercubes)

$\bar{X}_n$ is a **Markov chain** with transitions

$$\bar{\mathbf{P}}(k, l) = \frac{k}{d} 1_{k-1}(l) + \frac{d-k}{d} 1_{k+1}(l)$$



Observations:

- ▶ $\frac{d-k}{d} > \frac{k}{d}$ when $k < d/2 \Rightarrow$ The chain is more likely to move to the right.
- ▶ When $d/2 > k$, the chain is more likely to move back.
- ▶ The chain is **irreducible**, but **periodic** with period 2.

Ehrenfest model I

Example 12

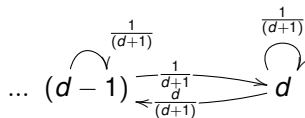
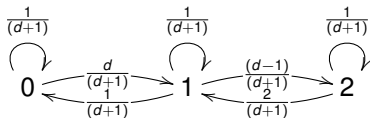
d balls are distributed in a couple of urns $\mathcal{U}, \mathcal{V}$. Select randomly a ball in one urn and place the ball in the other urn.

The number of balls B_n in $\mathcal{U}$ is a Markov chain ($S = \{1, \dots, d\}$)

$$\mathbb{P}(B_n = l \mid B_{n-1} = k) = \bar{\mathbf{P}}(k, l)$$

ϵ -lazy version ($\epsilon = \frac{1}{d+1}$)

$$\mathbf{P}_\epsilon(k, l) := \frac{1}{d+1} 1_k(l) + \frac{d-k}{d+1} 1_{k+1}(l) + \frac{k}{d+1} 1_{k-1}(l)$$



Ehrenfest model II

Example 12

Show that the invariant measure $\pi M = \pi = \pi M_\epsilon$ is

$$\pi(k) = \binom{d}{k} 2^{-d}$$

Let T be the first return time to $k \in S$. Calculate $\mathbb{E}(T | X_0 = k)$.

Applications:

- ▶ In 1907, Paul and Tatiana Ehrenfest introduced this model to analyze the thermodynamic equilibrium of gas molecules passing from one container into another through some hole.
- ▶ Birth and death model with a death rate B_n/b that depends on the size of the population.

Recurrent and transient states

Definition

Number of returns to x $N_x := \sum_{n \geq 1} 1_x(X_n) = \sum_{n \geq 1} 1_{X_n}(x)$

The potential matrix

$$\begin{aligned} G(x, y) &:= \mathbb{E} \left(\sum_{n \geq 0} 1_y(X_n) \mid X_0 = x \right) \\ &= 1_{x=y} + \mathbb{E}(N_y \mid X_0 = x) = \sum_{n \geq 0} M^n(x, y) \end{aligned}$$

Definition

x is said to be **recurrent** if the return probability to x is 1. Otherwise, the state is said to be **transient**.

A Markov chain is said **recurrent** when **all the states are recurrent**.

$T_{x,n}$: duration of the n -th excursion, a sequence of indep. r.v. with common law.

$$\mathbb{P}(N_x \geq n \mid X_0 = x)$$

$$= \mathbb{P}(T_{x,1} < \infty, T_{x,2} < \infty, \dots, T_{x,n} < \infty \mid X_0 = x)$$

$$= \mathbb{P}(T_{x,1} < \infty \mid X_0 = x)^n$$

$$\mathbb{E}(N_x \mid X_0 = x) = \sum_{q \geq 1} \mathbb{P}(T_{x,1} < \infty \mid X_0 = x)^q$$

Proposition 1

$$\begin{aligned} x \text{ is recurrent} &\iff \mathbb{E}(N_x \mid X_0 = x) = \infty \\ &\iff \mathbb{P}(T_{x,1} < \infty \mid X_0 = x) = 1 \iff G(x, x) = \infty \end{aligned}$$

For irreducible Markov chains, all the states have the same nature.

Proposition 2

$$y \text{ transient} \iff \mathbb{P}(T_{y,1} = \infty \mid X_0 = y) > 0$$

Proposition 3

For an irreducible chain,

$$\begin{aligned} y \text{ transient} &\iff \forall x \quad \mathbb{E}(N_y \mid X_0 = x) < \infty \\ &\iff \forall x \quad G(x, y) < \infty \end{aligned}$$

Proposition 4

For irreducible and recurrent Markov chains

$$\forall x, y \in S \quad \mathbb{P}(T_{y,1} < \infty | X_0 = x) = 1$$

Theorem 13

An irreducible chain is transient if and only if there is a non-trivial solution of

$$x_i = \sum_{j \neq i_0} p_{ij} x_j, \quad i \neq i_0,$$

$$0 \leq x_i \leq 1$$

where i_0 is an arbitrary state.

Kac's formula again

Theorem 14 (Kac's Theorem)

P: transitions of an **irreducible recurrent** Markov chain X_n .

T_x : the first return time to state $x \in S$.

The expected time spent in state y between visits to x

$$\gamma_x(y) := \mathbb{E} \left(\sum_{0 \leq n < T_x} 1_y(X_n) \mid X_0 = x \right)$$

is an **invariant measure** of the Markov chain $\gamma_x \mathbf{P} = \gamma_x$.

When the expected return times are finite,

$\mathbb{E}(T_x \mid X_0 = x) = \sum_y \gamma_x(y) < \infty$, the **unique invariant measure** of the chain is given by

$$\pi(x) = \frac{1}{\mathbb{E}(T_x \mid X_0 = x)}$$

Positive recurrent states

Definition

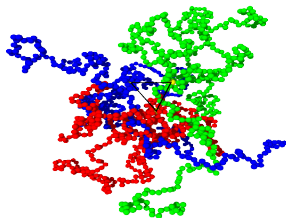
Recurrent states x s.t.

$$\mathbb{E}(T_x \mid X_0 = x) = \sum_{n \geq 1} \mathbb{P}(T_x \geq n \mid X_0 = x) < \infty$$

are called **positive recurrent states**. Otherwise [that is, $\mathbb{E}(T_x \mid X_0 = x) = \infty$], the recurrent state x is known as **null recurrent**.

An **irreducible** and **recurrent** Markov chain X_n has an **invariant probability measure** iff one of the states is **positive recurrent**. In this situation, all the states of the chain are **positive recurrent**.

Simple random walks (abbreviated SRW)



Countable state space model

$$x \in S = \mathbb{Z}^d$$

Any state $x \in \mathbb{Z}^d$ has $2d$ nearest neighbors denoted by $\mathcal{N}(x)$.

The simple random walk on the d -dimensional lattice $S = \mathbb{Z}^d$ is defined by the Markov transition

$$\mathbf{P}(x, y) = \frac{1}{2d} 1_{\mathcal{N}(x)}(y)$$

Starting from any state, we need an even number of steps to come back to the initial location.

Simple random walks

Dimension 1

X_n goes at each step randomly to the right or to the left with prob. $1/2$.

$\Rightarrow$ all states are **recurrent**

Dimension 2

$\Rightarrow$ all states are **recurrent**

Dimension $d \geq 3$

$\Rightarrow$ all states are **transient**